

Simulation-informed transfer learning improves low-data nanobody thermal stability prediction

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Experimental thermal-stability measurements for nanobodies are scarce, whereas simulation and physics-based calculations can produce many labels for related sequence perturbations. The practical question is how to use those labels when they do not measure the experimental endpoint itself. Here we study this problem for nanobody melting-temperature (T_m) prediction using NbBench as the experimental target dataset. We trained multi-task transfer models in which an ESM2 sequence encoder is shared by the T_m target task and computational source tasks, but all checkpoint selection, model selection, and final evaluation remain anchored to experimental T_m data. The source tasks included FEP-derived mutation free energies, Rosetta and ThermoMPNN mutation-effect scores, Rosetta scores on generated and random variants, and an MD-derived contact-persistence label. FEP labels gave the clearest improvement: held-out T_m mean absolute error decreased from 6.61 to 6.26°C relative to T_m-only training, with a paired improvement of 0.35°C (90% confidence interval, 0.24–0.46°C). The effect was not explained by simply using a larger ESM2 encoder, and it was not reproduced by the MD-derived contact-persistence label. Rosetta-scored generated variants gave a weaker but positive signal. These results show that simulation labels are useful for T_m prediction only when their physical content transfers to the experimental phenotype. In this case, mutation free energies appear to complement sparse absolute T_m measurements by providing local stability-change information around sequence variants.

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Introduction

Machine learning has made it possible to screen large molecular and materials spaces before costly experimental testing, but its usefulness still depends on the availability of reliable property labels. In materials informatics, this data bottleneck has motivated a simulation-to-real (Sim2Real) strategy: large computational databases generated by first-principles calculations, quantum chemistry, or molecular dynamics simulations are used as source data, and the resulting models are transferred to smaller experimental target datasets (1–3). A recent study in polymers and inorganic materials showed that the downstream prediction error of transferred models can scale with the amount of computational data, providing a quantitative way to judge whether a growing simulation database has real exper-

imental value (3). This framing is important because simulation data are not useful by default: simulations and experiments are separated by model assumptions, sampling limits, and systematic domain gaps.

Protein engineering faces an analogous problem, but with an additional layer of physical mismatch. Experimental measurements of protein properties are expensive and sparse, whereas biomolecular simulations and physics-based calculations can generate labels for many designed or mutated sequences. The challenge is sharper than in many property-matched transfer settings, because the simulated quantity may not be the same as the experimental target. For example, a target phenotype may be the melting temperature (T_m) of a protein, while a tractable simulation may report a mutation free-energy change, a structure-derived score, or a trajectory summary. The central question is therefore not simply whether simulation data can be added to a training set, but which simulated physical quantities can transfer to the experimental phenotype.

We examine this question using nanobody thermostability as a concrete case. Nanobodies, or single-domain VHH antibodies, are attractive engineering scaffolds because of their small size, modularity, and binding properties (4, 5). Their practical use, however, depends on thermal stability during expression, storage, formulation, and downstream screening. Experimental T_m measurements remain limited, and direct molecular simulation of thermal stability is computationally demanding (6). Sequence-based predictors and protein language models (PLMs) offer an attractive alternative: PLMs such as ESM2 learn broadly useful protein representations from large sequence corpora (7, 8), and recent methods have used deep sequence representations to predict protein or nanobody melting temperatures (9–11). Nevertheless, published benchmarks and related nanobody T_m studies indicate that target-specific supervision and careful validation remain essential (12–14).

Here we ask whether simulation-derived labels can provide that additional supervision when experimental T_m labels are scarce. We formulate T_m prediction as the target task and train source tasks from computational labels using a shared ESM2 sequence encoder. The source labels include FEP-derived mutation free energies, Rosetta and ThermoMPNN mutation-effect labels, Rosetta scores on generated and random variants, and

an MD-derived contact-persistence label. To keep the comparison target-centered, all model settings are selected using an experimental validation set, and all final claims are evaluated on the same held-out experimental Tm test set with paired error comparisons. This design prevents high performance on a source task from being mistaken for improved Tm prediction.

The main result is that the benefit of simulation-derived supervision is highly source-dependent. FEP mutation free-energy labels provide the strongest improvement over Tm-only training, whereas the MD-derived contact-persistence label does not improve held-out Tm prediction under the same evaluation design. Rosetta-scored generated variants provide a weaker but positive signal, suggesting a connection to future generator-physics-predictor design loops. Taken together, these results extend the Sim2Real argument from materials informatics to a biomolecular setting in which the source and target labels differ in physical meaning. Useful simulation data are not merely abundant; they encode a physical quantity that transfers to the experimental property of interest.

Materials and Methods

Experimental and technical design. This study was designed to test whether computational source labels can improve experimental nanobody melting-temperature (Tm) prediction when the source label is physically related to, but not identical with, Tm. We therefore used a multi-task transfer-learning design: experimental Tm prediction was the target task, and computational labels were source tasks that shared part of the neural network with the target task. The central constraint was that all model selection and all final claims were anchored to experimental Tm performance. Source-task losses were used for representation learning during training, but source-task validation errors were not used to select final checkpoints or final hyperparameters.

NbBench Tm data set and split definitions. Experimental Tm labels were taken from the thermo-seq task of NbBench, a nanobody benchmark data set with predefined splits (13). The processed files used in this study contained 57 training sequences, 114 validation sequences, and 396 test sequences. The terms were used as follows. The training split was used to fit model parameters. The validation split was used for checkpoint selection, early stopping, and hyperparameter selection. The test split was held out until the selected settings were fixed and was used only for final reporting. This definition applies to both Tm-only models and multi-task transfer models.

The target CSV files contained an amino-acid sequence column and a Tm label in degrees Celsius. During training, Tm labels were transformed to a scaled regression target. The scaler was fit only on the 57 training labels, using a robust scaling step followed by min-max

scaling to $[0,1]$. The fitted scaler was then applied to validation labels during training. At evaluation time, model predictions were inverse-transformed back to degrees Celsius before computing errors. The validation and test label distributions were therefore not used to fit the target-label scaling transform.

For the experimental-label scaling reference, the Tm training split was subsampled to the requested number of labels using the run seed. The validation and test splits were not subsampled. These experiments provide a target-label reference for interpreting whether computational source-label scaling behaves in the expected direction.

Computational source-label data sets. Computational labels were represented as source tasks rather than as additional Tm measurements. The source-label collection comprised FEP-derived mutation free energies, Rosetta mutation-effect scores, ThermoMPNN mutation-effect scores, Rosetta scores assigned to generated and random variants, and an MD-derived contact-persistence label. All source labels were preprocessed to $[0,1]$ -scaled regression targets and were used only through their own source-task heads.

Mutation-effect sources were organized as two template-specific tables derived from the 1MEL and 4IDL nanobody structures. FEP, Rosetta, and ThermoMPNN each contained 435 and 409 processed rows for the two templates. Rosetta-scored generated variants and Rosetta-scored random variants each contained 1000 and 1000 processed rows. The MD-derived contact-persistence source was represented by a single processed trajectory-summary table with 1143 rows. Row counts exclude headers.

For a mutation-effect experiment with requested source-label count n , up to n rows were sampled independently from each of the two template-specific source tables using the run seed. The two template-specific tables were assigned separate task identifiers. For an MD-derived source experiment with requested source-label count n , n rows were sampled from the single processed trajectory-summary table. After sampling, source-label rows were split 80/20 into source-training and source-validation rows using the same run seed. These source-validation rows were retained for monitoring and control analyses. In the reported final protocol, however, the validation data passed to the trainer for checkpoint selection and early stopping were filtered to the experimental Tm validation rows.

FEP mutation free-energy labels. The FEP source labels were mutation free-energy changes for variants of the 1MEL and 4IDL nanobody templates. FEP calculations followed the same protocol as our related nanobody thermostability study. Briefly, alchemical mutation simulations were run with NAMD 3 using GPU acceleration (15). Input systems were prepared in VMD (16) using a customized AlaScan workflow (17). Initial struc-

tures were obtained from AlphaFold 3 predictions (18). Systems used the CHARMM36 protein force field (19), TIP3P water (20), and 150 mM NaCl.

Each alchemical transformation used a dual-topology mutation setup with a hybrid-residue topology for CHARMM36. The transformation was divided into 20 equally spaced λ windows from wild type to mutant. Forward and backward transformations were simulated, with 1 ns per direction, a 2-fs integration timestep, and 200 ps equilibration per window before data collection. Simulations were performed in the NPT ensemble at 300 K and 1 atm with Langevin temperature control and a Langevin piston barostat. Bonds involving hydrogen atoms were constrained with SHAKE (21), and long-range electrostatics were evaluated with particle mesh Ewald (22). Relative free energies were estimated with the Bennett acceptance ratio (23). The processed learning labels used the sign convention and [0,1] scaling stored in the FEP CSV files.

Rosetta and ThermoMPNN mutation-effect labels. Rosetta mutation-effect labels were generated with the Rosetta ddg_monomer application (24). For each template, the PDB structure was cleaned to the target chain, variant sequences were compared with the corresponding wild-type sequence, and the sequence differences were converted to Rosetta mutation-list files. The calculation used full-atom Rosetta with the ref2015 score function (25), 50 iterations, Cartesian minimization enabled, local-only optimization disabled, side-chain-minimization-only disabled, minimum-score aggregation, and no PDB dumping. The same Rosetta protocol was used for the template mutation-effect tables, the ESM2-generated variants, and the random-variant control.

For the generated-variant source, ESM2 650M was used to sample 100,000 two-mutation variants from each wild-type template sequence, variants were ranked by pseudolog likelihood, and the top 1% were retained for Rosetta scoring (8). The random-variant control used approximately 1000 two-mutation variants generated by randomly choosing mutation positions and amino acid substitutions, followed by the same Rosetta scoring workflow.

ThermoMPNN source labels were computed from the same template-specific variant sets using ThermoMPNN, a stability-change predictor trained by transfer learning from larger protein-stability data (26). The ThermoMPNN outputs were converted to the same processed CSV format as the other mutation-effect source tasks.

Neural-network architecture. All models used an ESM2 protein language-model encoder (8) followed by lightweight regression heads. Unless otherwise stated, the encoder was facebook/esm2_t6_8M_UR50D. Sequences were tokenized with truncation and padding to a maximum length of 160 residues. The first-token en-

coder representation was passed to a shared multilayer perceptron with hidden dimensions 256, 128, and 32, ReLU activations, and dropout. A scalar Tm head predicted the scaled target Tm label. In source-transfer models, additional scalar source heads predicted the source-task labels from the same shared representation. The Tm-only baseline used the ESM2 encoder, the shared trunk, and the Tm head with only experimental Tm labels. Source-transfer models used the same target path plus source-task heads. The main FEP model used separate source heads for the two template-specific mutation-effect tasks, which was the best-performing source-head design in the source-head comparison. Additional controls evaluated a shared mutation-effect head, a shared head conditioned on a template embedding, and a shared head with template-specific affine calibration. Architecture controls also evaluated residual and latent source-fusion variants, but the main result used the original shared-trunk architecture.

Updated-encoder and frozen-encoder training. Two encoder-training regimes were evaluated. In the updated-encoder regime, the ESM2 encoder and regression heads were optimized jointly. The encoder used a separate learning rate from the newly initialized trunk and heads, so that the pretrained representation was updated more conservatively than the output layers. In the final FEP setting, the selected encoder learning rate was 3×10^{-5} and the head learning rate was the default 3×10^{-4} .

In the frozen-encoder control, all ESM2 parameters were fixed by setting their gradients off. The encoder was run once to precompute sequence embeddings for the target and source examples, and training then optimized only the downstream trunk and task heads. This control asks whether the source labels help when the pretrained sequence representation is used purely as a fixed feature extractor. Encoder-size controls repeated selected Tm-only and FEP-assisted settings with 35M- and 650M-parameter ESM2 encoders.

Loss function and multi-task objective. Each training example was represented as (x_i, y_i, t_i) , where x_i is the sequence, y_i is the scaled regression label, and t_i is the task identifier. The target Tm task used one task identifier, and each source task used its own identifier. A minibatch could contain target and source examples, but each example contributed loss only through the head for its task.

The per-task regression loss was Huber loss with $\delta = 1.0$ on scaled labels. The default multi-task objective used learnable task-uncertainty weights (27). For task k with loss L_k and learned log-scale s_k , the contribution was

$$\frac{L_k}{2\exp(2s_k)} + s_k.$$

The total loss was the sum over the tasks present in the minibatch. Fixed source-task weights were also evalu-

ated as controls, particularly for the MD-derived source labels.

Training, early stopping, and hyperparameter selection. Models were trained with AdamW, cosine learning-rate scheduling, batch size 16, weight decay 0.04, 100 warmup steps, and at most 400 epochs. The default head learning rate was 3×10^{-4} , the default updated-encoder learning rate was 1×10^{-4} , and the default dropout rate was 0.15. Training used mixed precision on CUDA devices. Checkpoints were saved and evaluated once per epoch. Within each run, the best checkpoint was the checkpoint with the lowest mean squared error on the experimental Tm validation examples. Early stopping used the same validation metric with a patience of 15 epochs. In source-transfer models, source-label rows were present in the training set, but they were removed from the validation data passed to the trainer for checkpoint selection and early stopping. This is the implementation-level step that ensures that the reported checkpoints are selected by target-task validation performance rather than by source-task performance.

Hyperparameter selection was also performed on the experimental Tm validation split. Candidate settings varied encoder learning rate, head learning rate, dropout, loss-weighting mode, source-head design, and selected architecture controls. The search included encoder learning rates of 3×10^{-5} , 1×10^{-4} , and 3×10^{-4} ; a lower head learning rate of 1×10^{-4} paired with encoder learning rate 3×10^{-5} ; dropout rates of 0.05, 0.15, and 0.30; and fixed source-task loss weights where relevant. Each candidate setting was evaluated as a three-seed ensemble on the Tm validation split. The selected setting for each source class was then evaluated on the held-out test split as a ten-seed ensemble.

Independent run seeds controlled model initialization, target-training subsampling in the target-label scaling experiment, source-label sampling, and source-task train/validation splitting. Final comparisons used seeds 1–10. The label-count curves used fixed source-specific training recipes while varying the number of source labels, rather than retuning a separate model for every label-count point.

Final evaluation and statistics. Final evaluation used the same 396 held-out Tm test examples for all conditions. For each condition, the ten independently seeded models produced scaled Tm predictions for every test sequence. Predictions were averaged in scaled-label space and then inverse-transformed to degrees Celsius. The primary metric was mean absolute error (MAE) in degrees Celsius. Root mean squared error, coefficient of determination, Spearman correlation, and Pearson correlation were also computed, but MAE was used for the main comparisons because it is directly interpretable as average temperature error.

Confidence intervals were estimated by nonparametric bootstrap resampling over held-out test proteins. For a

single condition, the absolute-error vector from the ten-seed ensemble was resampled with replacement 10,000 times using a fixed random seed. The reported 90% confidence interval is the 5th to 95th percentile range of the bootstrapped MAE values.

For paired comparisons, the same bootstrap indices were applied to the absolute-error vectors from the candidate and reference conditions. The candidate-minus-reference MAE difference was then computed for each bootstrap sample. This paired bootstrap preserves the matched-test-example structure of the comparison. In figures and result tables, Δ MAE is candidate minus reference, so negative values indicate lower error than the reference. The one-sided tail probability stored in the result JSON is the fraction of paired bootstrap samples in which candidate-minus-reference Δ MAE is greater than zero. The manuscript emphasizes effect sizes and 90% confidence intervals.

Implementation details. The training and evaluation pipeline is implemented in `prepare.py` and `train.py`. The data loader constructs target and source tasks, applies target-label scaling, performs seeded source-label sampling, filters validation data for target-task checkpoint selection, evaluates seed ensembles, and writes bootstrap statistics. Each result directory contains a machine-readable `scaling.json` file with command-line arguments, resolved hyperparameters, per-point metrics, bootstrap intervals, and paired comparisons.

Results

A target-centered framework for using simulation labels in Tm prediction. We first established an evaluation framework for asking whether simulation-derived labels improve experimental Tm prediction. The target task was nanobody Tm regression from sequence. Computational labels were treated as source tasks rather than as substitutes for experimental Tm. Each model shared an ESM2 sequence encoder between a Tm prediction head and a source source-label head, allowing the source task to shape the representation while keeping the final objective anchored to experimental Tm prediction (Fig. 1). The comparison was deliberately target-centered. All candidate settings were selected using an experimental Tm validation set, and the final numbers were computed only on a held-out experimental Tm test set. Improvements were assessed by paired differences in absolute error on the same test examples. This design addresses a common ambiguity in simulation-to-experiment learning: a source task can be learnable and physically meaningful without improving the target experimental phenotype.

Source-label scaling depends on the physical label. We next varied the number of experimental and computational labels to determine whether more source data translated into better Tm prediction. As expected,

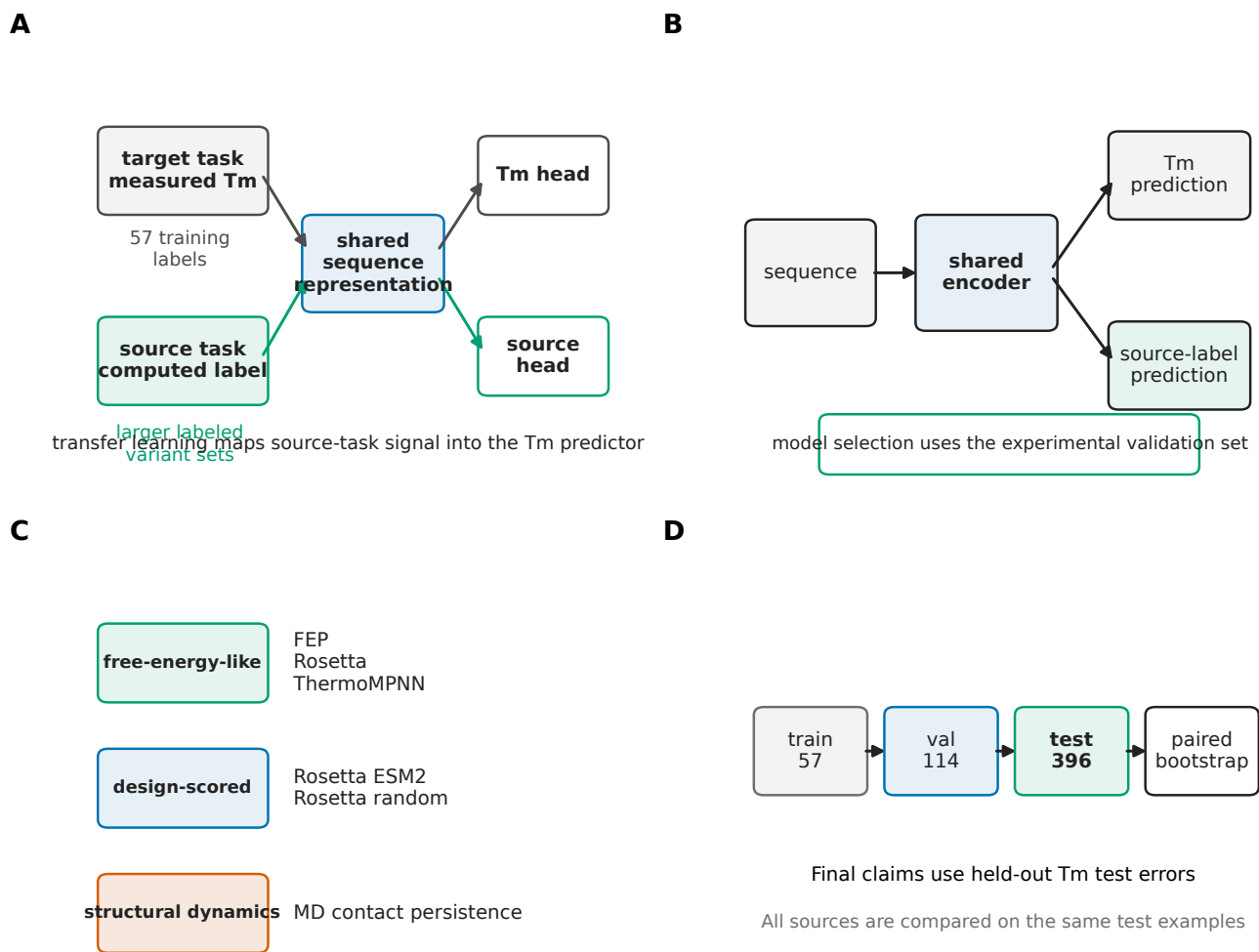


Fig. 1. Multi-task transfer-learning framework for using simulation data in experimental Tm prediction. (a) Experimental Tm labels are scarce, whereas simulation and physics-based calculations can provide larger source-label pools. (b) Nanobody Tm prediction is used as a concrete case in which the source labels are related to, but not identical with, the experimental target. (c) The model shares an ESM2 sequence encoder between the Tm target head and a source-label head. (d) Model settings are selected on an experimental validation set, and final claims are evaluated on held-out experimental Tm test examples using paired error statistics.

increasing the number of experimental Tm labels improved Tm-only performance (Fig. 2). FEP mutation free-energy labels also showed a beneficial trend: the largest FEP label-count setting tested gave the best held-out Tm error among the FEP settings. The curve was not strictly monotonic, so we do not interpret it as a clean power law. Instead, we interpret it as evidence that additional FEP labels can improve the target task when they are selected and evaluated through the experimental Tm endpoint.

The MD-derived contact-persistence label behaved differently. Increasing the number of these labels did not yield a robust improvement over Tm-only learning, even though the source labels were also generated from simulations. Thus, the label-count experiment already suggested the central result of the paper: source-label quantity is not sufficient. The transferable physical content of the source label matters.

FEP mutation free-energy labels provide the strongest final improvement. We then compared the selected settings for each source label. Tm-only training achieved a held-out test MAE of 6.61°C. Adding FEP mutation free-energy labels reduced the MAE to 6.26°C, corresponding to a paired improvement of 0.35°C (90% confidence interval, 0.24–0.46°C; Fig. 3). This was the clearest improvement observed across all source labels.

The other sources showed weaker or negative effects. Rosetta-scored variants generated by ESM2 improved over Tm-only learning (paired improvement, 0.17°C; 90% confidence interval, 0.03–0.31°C), and ThermoMPNN mutation-effect labels showed a similar but smaller gain. Rosetta mutation scores and Rosetta-scored random variants were weaker and had confidence intervals that crossed zero. The MD-derived contact-persistence label worsened the final test error relative to Tm-only learning. These results support a selective, source-dependent interpretation: mutation free-energy-

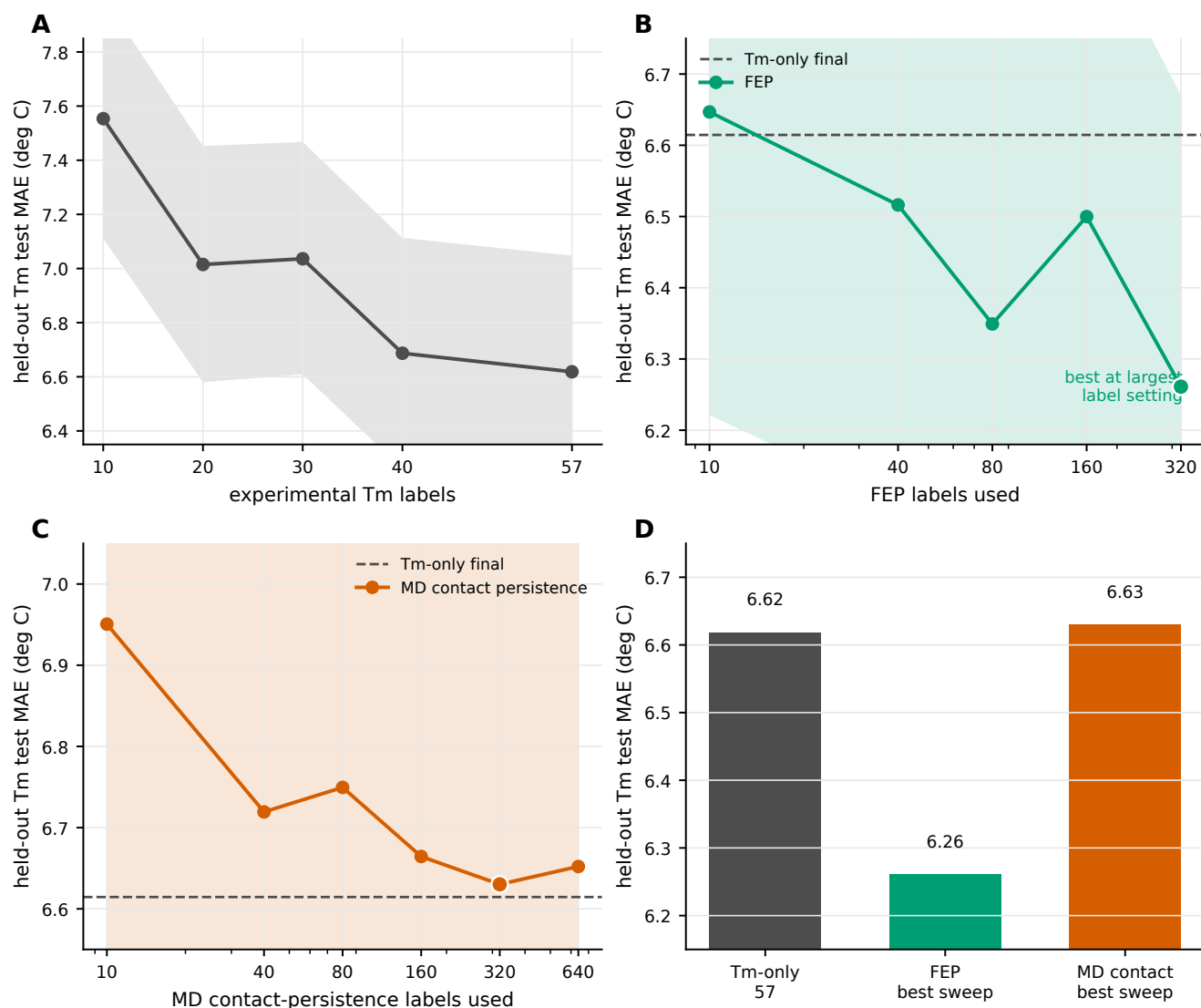


Fig. 2. Label-count scaling of experimental and computational supervision. (a) Tm-only learning improves as more experimental Tm labels are used. (b) FEP mutation free-energy labels reach their best held-out Tm error at the largest source-label setting tested, although the curve is not strictly monotonic. (c) The MD-derived contact-persistence label does not show a robust improvement over Tm-only learning. (d) Best points from the label-count sweeps separate FEP from the Tm-only and MD-derived contact-persistence references.

like labels were useful, but a structural trajectory summary was not sufficient for this target task.

The FEP gain is not explained by encoder size alone. Because PLM scale can affect downstream performance, we tested whether the benefit of FEP labels could be reproduced simply by increasing the ESM2 encoder size. It could not. In the Tm-only setting, larger 35M and 650M encoders did not improve held-out performance relative to the 8M encoder. In the FEP setting, the 35M and 650M encoders reached 6.50 and 6.65°C MAE, respectively, both worse than the 8M FEP model. FEP labels remained beneficial within each size comparison, but the best absolute performance was obtained with the smallest encoder and FEP source labels. Thus, in this low-Tm-data regime, the dominant gain came from the physical source label rather than from scaling the

language model.

Encoder adaptation helps exploit the source-label signal. We also asked whether the FEP gain required updating the sequence encoder. In a frozen-encoder control, overall errors increased, but FEP still improved over the frozen Tm-only baseline: the frozen Tm-only model achieved 7.38°C MAE, whereas frozen FEP achieved 7.20°C MAE. The improvement was smaller than in the updated-encoder setting, and the frozen FEP model remained worse than the updated-encoder FEP model. These results indicate that FEP labels carry useful information even when the representation is fixed, but limited encoder adaptation is important for extracting the strongest target-task benefit.

Physics-labeled generated variants connect transfer learning to design loops. Rosetta labels on ESM2-

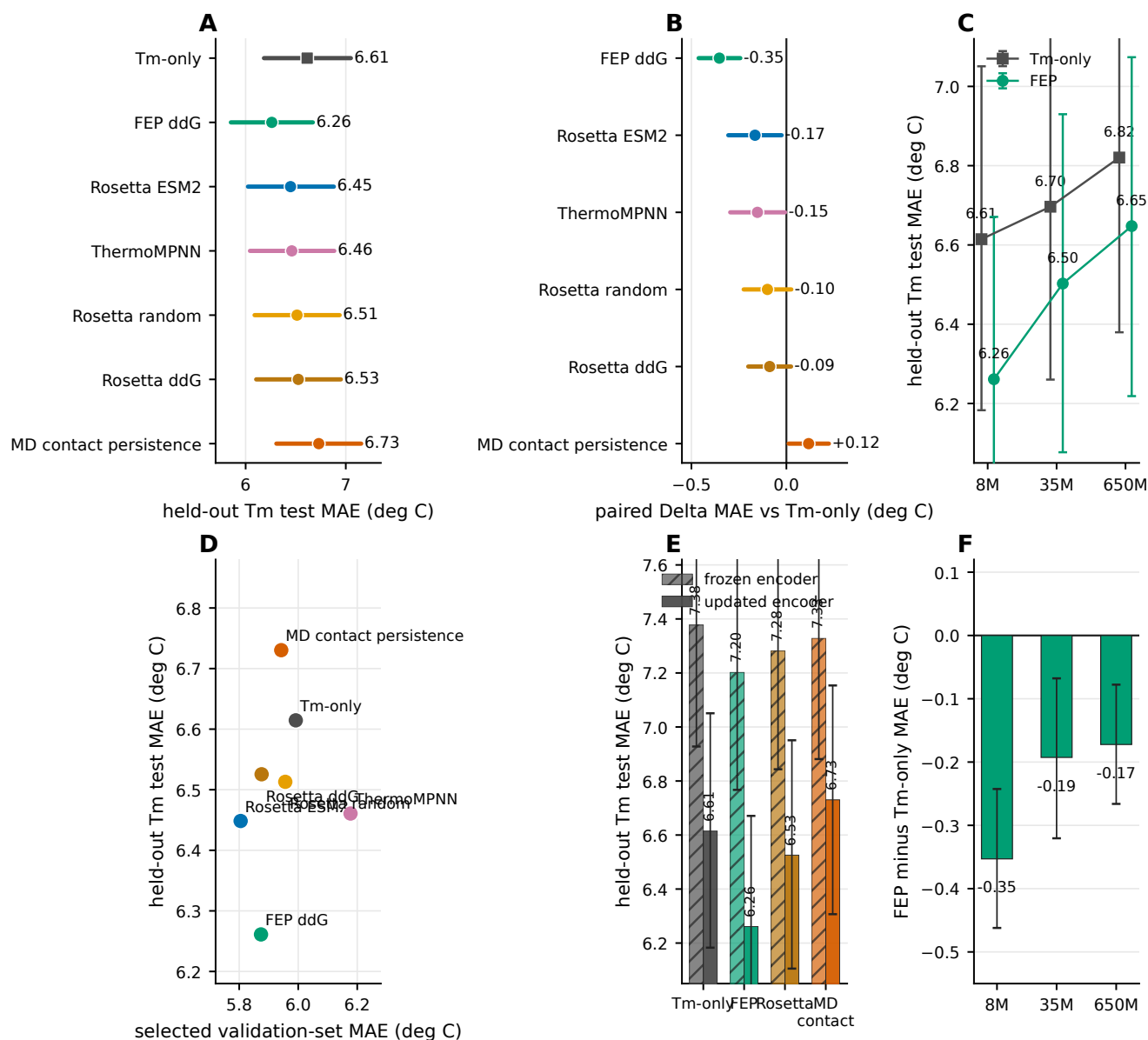


Fig. 3. Final held-out Tm performance across source labels and controls. (a) Final test MAE for Tm-only training and each source label. Intervals show bootstrap confidence intervals over test-set absolute errors. (b) Paired Δ MAE relative to Tm-only training. Negative values indicate improvement. (c) ESM2 size controls show that larger encoders do not improve the selected Tm-only or FEP settings. (d) Validation-set performance and held-out test performance are related but not identical, emphasizing the need for final target-task testing. (e) Frozen and updated-encoder comparisons show that FEP helps in both regimes, with the strongest absolute performance obtained when the sequence representation is updated. (f) Paired FEP gains remain negative across ESM2 sizes, but the best absolute error is obtained with the 8M encoder.

proposed variants improved Tm prediction over the Tm-only baseline, whereas the comparison between ESM2-proposed and random Rosetta-scored variants was suggestive but not conclusive. The paired improvement of ESM2-proposed variants over Tm-only learning was 0.17°C (90% confidence interval, $0.03\text{--}0.31^{\circ}\text{C}$). By contrast, the paired difference between ESM2-proposed and random variants was 0.07°C in favor of ESM2-proposed variants, with a confidence interval crossing zero. We therefore interpret this result conservatively. Physics-labeled generated variants retain useful supervised signal for Tm prediction, but the present data do not prove that the current generator is superior to random variant

sampling.

Mutation free-energy labels provide a physical interpretation. The final pattern suggests why FEP labels transfer to Tm prediction even though they are not Tm measurements. Experimental Tm labels provide sparse anchors for the absolute thermal-stability phenotype. Mutation free-energy labels provide local information about how stability changes under sequence perturbations. When learned jointly, these two views can shape a representation that is more useful for held-out Tm prediction than either sparse Tm labels alone or a generic simulation-derived summary. The negative result for the

MD-derived contact-persistence label defines the boundary of this interpretation: a simulation label must encode the right physical relation to the target phenotype.

Discussion

This study asked how simulation-derived labels should be used when the experimental target is scarce and the simulated quantity is related to, but not identical with, that target. Using nanobody T_m prediction as a case study, we found that FEP mutation free-energy labels improved held-out experimental T_m prediction more clearly than any other source label tested. The result was not reproduced by a simple MD-derived contact-persistence label, nor was it explained by using a larger ESM2 encoder. These findings support a selective view of Sim2Real learning for biomolecules: simulation data are useful when the source label encodes physical information that transfers to the experimental phenotype.

The main conceptual point is that the helpful source label did not measure T_m itself. FEP estimates mutation free-energy changes, whereas T_m is an experimental phenotype reflecting the temperature at which folded and unfolded states are balanced under a particular assay condition. These are not interchangeable quantities. Nevertheless, both describe the same underlying stability landscape from different perspectives. Sparse T_m measurements anchor absolute thermal-stability values for observed sequences, while mutation free-energy labels provide local directions in sequence space. The improvement from FEP therefore suggests that source labels can help even when they are not direct surrogates for the target, provided that they describe a physically relevant perturbation of the target property.

This result extends the materials-informatics Sim2Real argument to a more heterogeneous biomolecular setting. In materials applications, transfer learning often links computational and experimental estimates of related material properties, and scaling behavior can quantify the downstream value of a growing computational database (3). In the present protein case, the most useful computational labels were not additional T_m measurements but mutation free energies. The label-count sweep for FEP was consistent with a beneficial scaling trend but was not smooth enough to claim a universal biomolecular scaling law. The stronger conclusion is source dependence: simulation-derived data should be evaluated by target-task performance, and source labels must be chosen for transferable physical content rather than availability alone.

The encoder controls sharpen this conclusion. Prior work on PLMs has shown that fine-tuning can be critical for downstream protein-property prediction (14), and related nanobody T_m analyses suggest that larger pretrained ESM2 models do not automatically produce better T_m predictors. Our own size controls reached the same practical conclusion: 35M and 650M ESM2 encoders did not outperform the 8M encoder in either

T_m-only or FEP-assisted settings. The best model used the 8M encoder with FEP source labels. Thus, for this dataset, the bottleneck was not simply language-model capacity. The more important ingredient was additional supervision carrying information about sequence-dependent stability changes.

The frozen-encoder experiment further indicates how the source-label signal is used. FEP labels improved performance even when the encoder was fixed, showing that part of the signal can be captured by the existing sequence representation. However, the best FEP model required updating the encoder. This suggests that FEP labels help most when they are allowed to reshape the representation toward stability-relevant sequence directions while still being selected by experimental T_m performance. In practical terms, the result argues for controlled encoder adaptation rather than either fully fixed feature extraction or unconstrained source-task optimization.

The negative result for the MD-derived contact-persistence label is equally important. It shows that the failure mode is not a lack of simulation in general, but a mismatch between the source label and the target phenotype. Contact persistence in high-temperature trajectories may report aspects of structural durability, but it is a compressed trajectory statistic and may miss the mutation-specific energetic effects that determine how sequence changes alter the stability landscape. This distinction matters for future database design. A simulation campaign optimized for target transfer should not only maximize the number of labels; it should choose labels whose physical meaning is expected to transfer to the experimental decision being made.

The generated-variant result points toward a possible design workflow. Rosetta scores assigned to ESM2-generated variants improved T_m prediction relative to T_m-only learning, although generated variants were not conclusively better than random variants in the current comparison. This should be viewed as a bridge, not a completed design loop. The positive signal indicates that proposed sequences can be labeled by physics-based calculations and used as supervised source data for a T_m predictor. A natural next step is an iterative loop in which a generator proposes variants, physics-based calculations annotate them, the T_m predictor is updated, and candidates are selected for experimental testing. The present work supplies evidence for the supervised learning component of that loop, but it does not test active learning, reinforcement learning, or experimental validation of newly proposed high-T_m nanobodies.

Limitations. Several limitations should guide interpretation. First, the experimental T_m training set is small, and the conclusions should be tested on additional nanobody collections and other protein families. Second, the source-label comparison is limited to the computational labels available here. Other free-energy protocols, stability predictors, trajectory summaries, or structure-

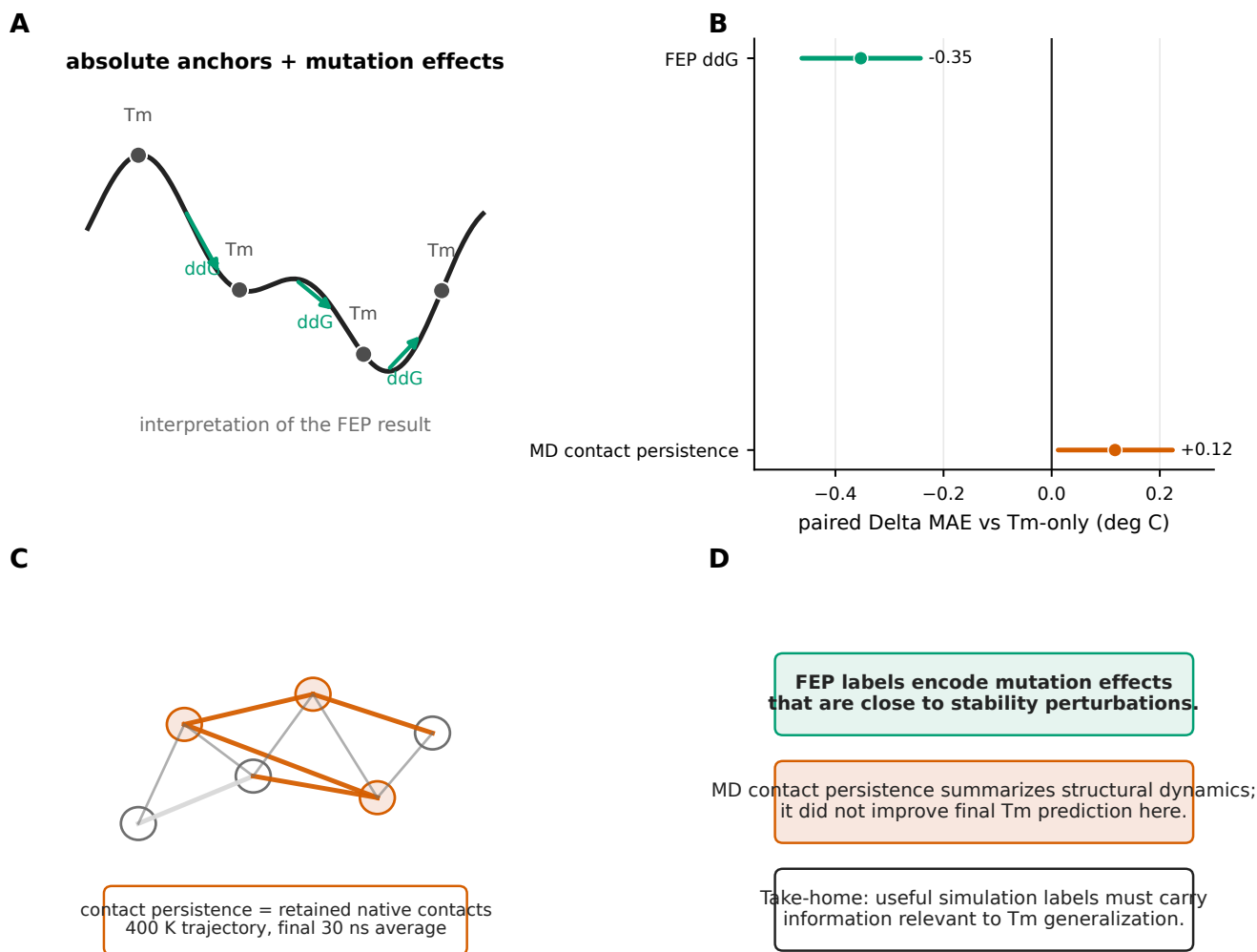


Fig. 4. Interpretation and boundary condition. (a) Experimental T_m labels provide sparse absolute stability anchors, whereas mutation free-energy labels provide local stability-change directions. (b) FEP improves T_m prediction, whereas the MD-derived contact-persistence label does not. (c) Contact persistence summarizes retained native contacts in high-temperature trajectories, but this structural-dynamics signal does not necessarily transfer to held-out T_m prediction. (d) The useful signal is not generic simulation augmentation, but source-label information with transferable physical content.

based descriptors may transfer differently. Third, the FEP curve does not establish a clean power-law scaling relation; it supports a beneficial trend and a strong final improvement, but not a universal scaling exponent. Fourth, no new experimental measurements were performed to validate high-stability candidates. Finally, raw simulation trajectories may be too large to deposit fully, so reproducibility will depend on processed data, scripts, and clear reporting of the simulation protocol.

Outlook. The practical lesson is that simulation labels should be treated as design choices. For low-data experimental protein-property prediction, the relevant question is not whether simulations can produce many labels, but whether those labels encode information that improves the target task under a target-centered validation protocol. In this study, mutation free-energy labels provided that information for nanobody T_m pre-

diction. This supports a broader strategy for biomolecular design: use experimental data to define the target phenotype, use simulation to annotate physically meaningful perturbations around that phenotype, and evaluate the combined model only on held-out experimental data. Such a strategy offers a disciplined path from simulation-assisted prediction to future closed-loop protein engineering.

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AUTHOR CONTRIBUTIONS

Yasuhiro Matsunaga conceived and supervised the study, acquired funding, developed the analysis and training code, performed the analyses, interpreted the results, prepared the figures, and wrote the manuscript. Kentaro Sasaki performed computational calculations.

Taihei Murakami contributed to draft writing. All authors reviewed and approved the final manuscript.

Declaration of competing interest

Taihei Murakami is an employee of Epsilon Molecular Engineering, Inc. Yasuhiro Matsunaga is a scientific advisor of Epsilon Molecular Engineering, Inc.

Data availability

Processed data required to reproduce the main and supplementary figures will be deposited in Zenodo at [ZENODO_DATA_DOI_PLACEHOLDER](#). Raw simulation trajectories are large and will be made available at [RAW_DATA_LOCATION_OR_REQUEST_POLICY_PLACEHOLDER](#).

Code availability

Analysis, training, and figure-generation code will be available on GitHub at [GITHUB_REPOSITORY_URL_PLACEHOLDER](#). An archival release will be deposited in Zenodo at [ZENODO_CODE_DOI_PLACEHOLDER](#).

AI-use disclosure

AI-assisted tools were used to support language editing, code development, and figure preparation. The authors reviewed and take responsibility for all manuscript content, analyses, and conclusions.

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Supplementary Note 1: Supplementary Information

A. Reproducibility checklist. All main comparisons used the same experimental Tm endpoint and the same held-out test set. The experimental split contained 57 training sequences, 114 validation sequences, and 396 held-out test sequences. The validation split was used for checkpoint selection, early stopping, and hyperparameter selection. The test split was used only for final reporting after the selected settings were fixed. Each final comparison used ten independently seeded models, averaged as an ensemble before test errors were computed. Validation-set searches used three independently seeded models per candidate setting.

The exact run arguments and resolved hyperparameters are stored in the scaling.json files under each result directory. These JSON files are the primary machine-readable record for the model settings used in the figures. The pipeline scripts read processed CSV files, build the target and source tasks, train the models, evaluate the seed ensembles, and write bootstrap statistics without manual postprocessing of prediction files.

B. Data splits and source-task definitions. Table 1 summarizes the processed data used by the reported comparisons. Row counts exclude header rows. Mutation-effect sources were represented by two template-specific source tasks; a setting with n labels therefore sampled up to n labels from each of the two source tables. For example, the final FEP setting used $n = 320$ from each template-specific table before the 80/20 source-task split. The MD-derived contact-persistence source was represented by a single processed trajectory-summary table.

Table 1. Processed target and source-label data.

Data type	Role	Processed rows	Use in reported comparisons
Experimental Tm	Target train	57	Model fitting
Experimental Tm	Target validation	114	Checkpoint and setting selection
Experimental Tm	Target test	396	Final held-out evaluation
FEP mutation free energy	Source tasks	435 + 409	Main source task
Rosetta mutation score	Source tasks	435 + 409	Source comparison
ThermoMPNN mutation score	Source tasks	435 + 409	Source comparison
Rosetta-scored random variants	Source tasks	1000 + 1000	Source comparison
Rosetta-scored generated variants	Source tasks	1000 + 1000	Source comparison
MD-derived contact persistence	Source task	1143	Source comparison and scaling control

Experimental Tm labels were read from the fixed NbBench train, validation, and test CSV files. Tm scaling was fit only on the target training labels. Source labels were read from precomputed [0,1]-scaled columns in the processed source CSV files. Within each seeded run, source rows were sampled using the run seed and then split 80/20 into source-training and source-validation rows. The target validation rows were not resampled.

C. Model-selection workflow. The model-selection workflow separated three decisions: within-run checkpoint selection, hyperparameter selection, and held-out testing. Within each training run, checkpoints were selected by the lowest validation-set mean squared error on experimental Tm rows. Early stopping also monitored only these experimental Tm validation rows. This prevented a model from being chosen because it predicted source labels well while failing to improve the target task.

Hyperparameter selection was performed on the experimental Tm validation split. For each source class, candidate settings included the default learning rates, lower and higher encoder learning rates, a lower head learning rate combined with a lower encoder learning rate, dropout rates of 0.05 and 0.30 in addition to the default 0.15, and alternative source-loss weighting where relevant. Architecture controls included the shared-head model used in the main comparison and residual or latent source-fusion variants. Mutation-effect head controls compared separate template-specific heads, a shared head, a context-conditioned shared head, and a calibrated shared head. The final main comparison used the best validation-set setting within each source class and then evaluated that selected setting on the held-out test split.

The label-count curves and the final source comparison answer different questions. The label-count curves hold a source-specific training protocol fixed while varying the number of available source labels, showing how the target error changes as source-label count increases. The final source comparison selects one validation-set setting per source and compares their held-out test errors against the selected Tm-only reference.

D. Training details. The default encoder was facebook/esm2_t6_8M_UR50D. Sequences were tokenized with truncation and padding to length 160. The first token representation was passed through a multilayer perceptron with hidden dimensions 256, 128, and 32, ReLU activations, and dropout. The final main architecture used this shared trunk with a Tm head and source-specific source heads. The loss for each task was Huber loss with $\delta = 1.0$ on scaled

labels. The default multi-task objective used learnable task-uncertainty weights; fixed weights were included in the search grid for source classes where loss balance was a plausible confounder.

The maximum training length was 400 epochs with batch size 16, AdamW, weight decay 0.04, a cosine learning-rate schedule, and 100 warmup steps. The default head learning rate was 3×10^{-4} . In updated-encoder models, encoder parameters were trained with their own learning rate, selected from the validation-set search. In frozen-encoder controls, ESM2 parameters were not updated and embeddings were precomputed before head training.

E. Statistical reporting. The primary reported statistic is MAE on the 396 held-out test examples after averaging predictions from ten seeded models. Confidence intervals were computed by resampling test examples, not by treating training seeds as independent experimental replicates. For single-condition MAE intervals, 10,000 bootstrap samples were drawn from the absolute-error vector and the 5th and 95th percentiles were reported. For paired comparisons, both conditions were resampled with the same bootstrap indices and the candidate-minus-reference MAE difference was computed for each resample. This paired procedure preserves the fact that all models are evaluated on the same test proteins.

The one-sided bootstrap tail probability recorded in the result JSON is the fraction of paired bootstrap samples in which candidate-minus-reference Δ MAE is greater than zero. It is reported as a diagnostic for the direction of the paired difference, but the main text emphasizes the effect size and the 90% confidence interval.

F. Machine-readable result files. The main source comparison is summarized by results/source_screen/final_source_screen_summary.json. The label-count curves for experimental T_m labels, FEP mutation free energies, and MD-derived contact persistence are stored as scaling.json files in the corresponding result directories and are referenced by the figure-generation script. Source-head controls are summarized by results/ddg_head_search/final_ddg_head_summary.json. Additional source-combination controls are summarized by results/abcd_search/final_abcd_summary.json. These files record the run arguments, selected settings, per-condition MAE values, confidence intervals, and paired bootstrap comparisons used to generate the figures.

G. Interpretation of the scaling curves. The experimental T_m label-count curve is an internal reference showing that the pipeline can detect the expected improvement when more target labels are added. The FEP curve evaluates whether increasing a physically related mutation-free-energy source improves the same held-out T_m endpoint. The FEP curve is beneficial but not strictly monotonic, so it should not be interpreted as evidence for a universal power law. The MD-derived contact-persistence curve shows that increasing the amount of a simulation-derived label is not sufficient by itself. The relevant conclusion is source dependence: target-task transfer depends on whether the source label encodes physical information that is useful for experimental T_m prediction.